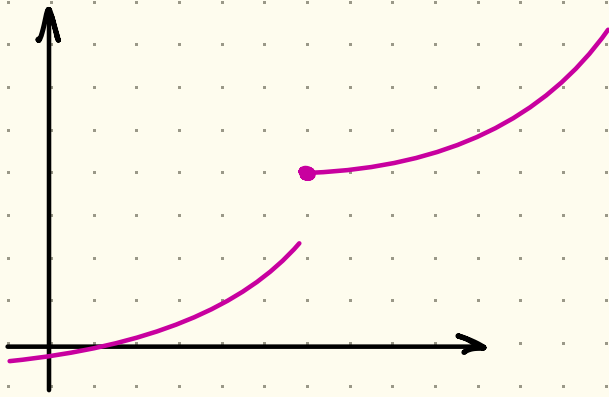
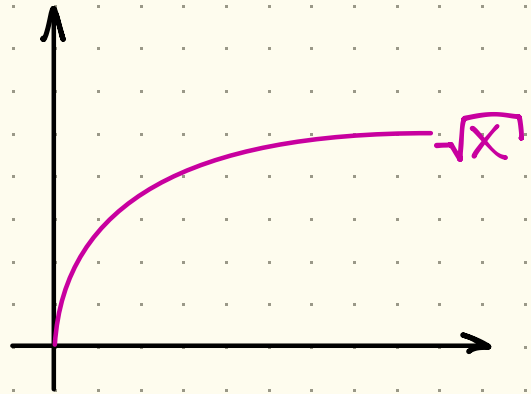


FUNZIONI NON DERIVABILI

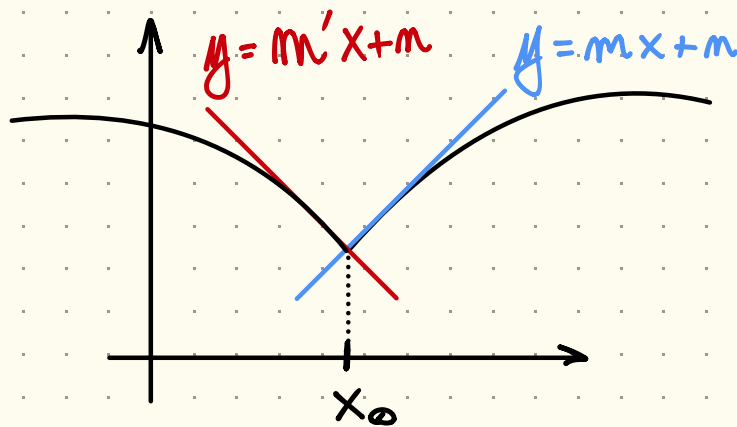
ESEMPIO.



NON DERIVABILE
POICHÉ NON CONTINUA



NON DERIVABILE
POICHÉ IN 0 LA
TANGENTE È VERTICALE



$$\lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0} = m' \neq m = \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0}$$



NON DERIVABILE IN x_0

es)

$$f(x) = \begin{cases} \sin x & x < 0 \\ 1 & x \geq 0 \end{cases}$$

f È DERIVABILE IN 0?

$$\bullet \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{\sin x}{x} = 1$$

$$f(0) = 1$$

$$\bullet \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 1 = 1$$

$$\bullet \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{\frac{\sin x}{x} - 1}{x} = \lim_{x \rightarrow 0^-} \frac{\sin x - x}{x^2}$$

$$\stackrel{H}{=} \lim_{x \rightarrow 0^-} \frac{\cos x - 1}{2x} \stackrel{H}{=} \lim_{x \rightarrow 0^-} \frac{-\sin x}{2} = 0$$

QUINDI $f'_-(0) = 0$

$$\bullet \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \frac{1 - 1}{x} = 0$$

QUINDI $f'_+(0) = 0$

$$\bullet \exists f'(0) = 0$$

WEIERSTRASS (TEOREMA)

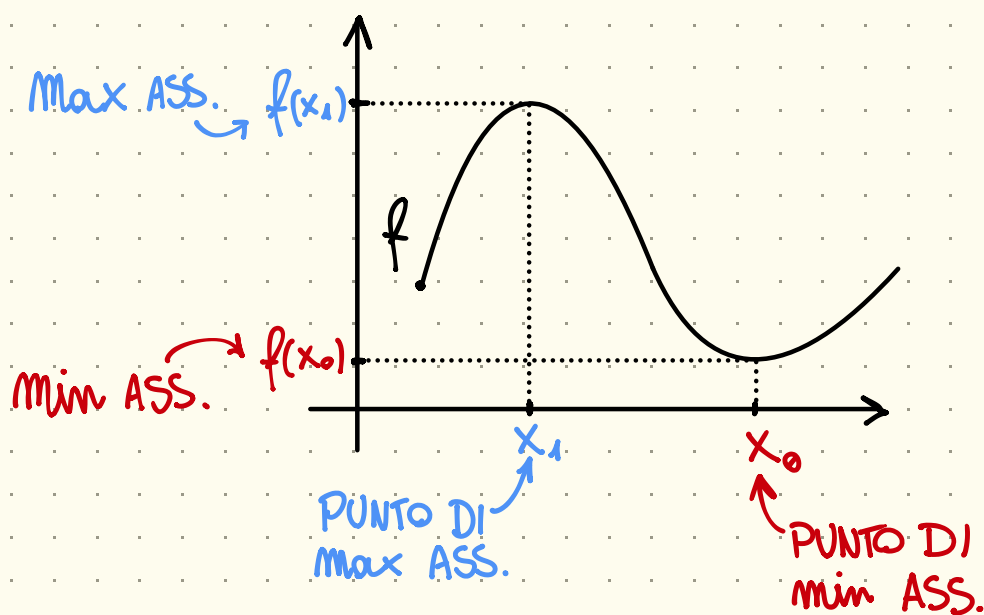
• $f[a, b] \rightarrow \mathbb{R}$ CONTINUA

ALLORA f AMMETTE **max** E **min** ASSOLUTI



$\exists x_0, x_1 \in [a, b]$ T.C.

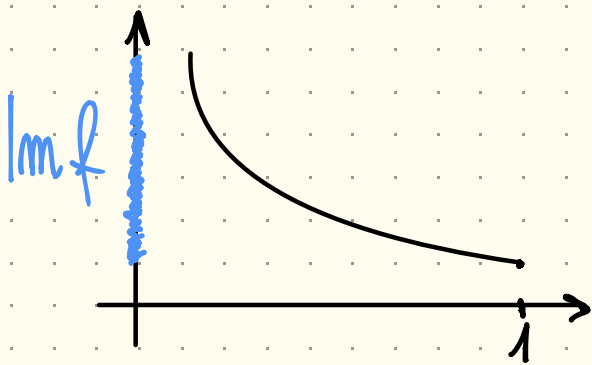
$f(x_0) \leq f(x) \leq f(x_1) \quad \forall x \in [a, b]$



VALORI INTERMEDI (TEOREMA)

$f: I \rightarrow \mathbb{R}$ CONTINUA, I INTERVALLO

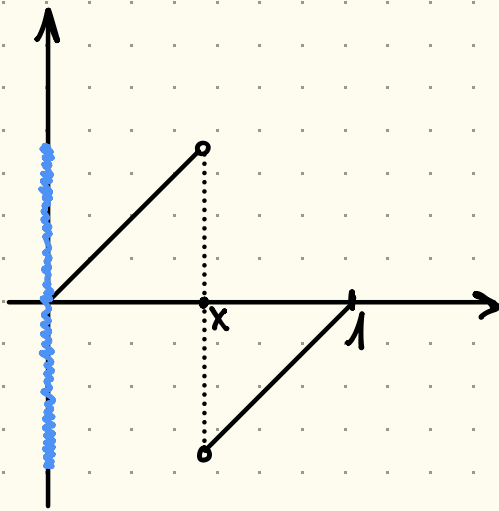
ALLORA $\text{Im } f$ É UN INTERVALLO



- $f(0, 1] \rightarrow \mathbb{R}$

- $f(x) = \frac{1}{x}$

- $\nexists \max$



- $f[0, 1] \rightarrow \mathbb{R}$

- f NON CONTINUA

es) $f: [0, 1] \rightarrow \mathbb{R}$

- $f(x) = \frac{x + \cos(10x)}{\sqrt{x+1}}$

• f CONTINUA, $\text{Dom } f = [0, 1]$

$\Downarrow$ ^W

$\exists x, x_1 \in [0, 1]$ T.C.

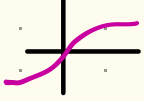
$$f(x_0) \leq f(x) \leq f(x_1) \quad \forall x \in [0, 1]$$

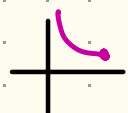
$\uparrow$
PUNTO DI max

$\leftarrow$ PUNTO DI min

• $f(x) = \pi \quad \forall x \in [0, 1] \xRightarrow{\text{VI}} \text{Im } f \text{ É UN INTERVALLO}$

4 POSSIBILITA'

• $(\text{Im } f, \sup f)$ 

• $[\text{Im } f, \sup f)$ 

$$\text{Im } f = (\text{Im } f, \sup f]$$

$[\text{Im } f, \sup f]$ 

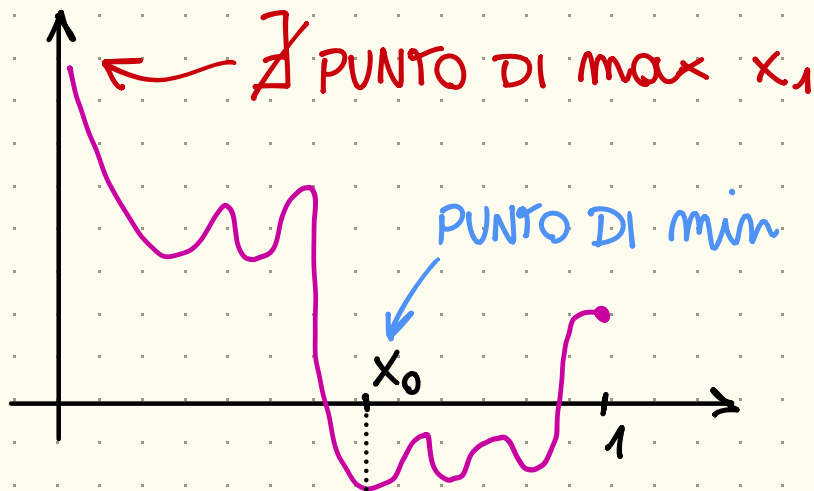
es)

$$f: (0, 1] \rightarrow \mathbb{R}, f(x) = \frac{\cos x}{x}$$

$\text{Dom } f = (0, 1] \rightarrow \text{NON POSSO APPLICARE W}$

• f CONTINUA

• $\lim_{x \rightarrow 0^+} \frac{\cos x}{x} = +\infty$



VI $\Rightarrow \text{Im } f$ É UN INTERVALLO

ZERI (TEOREMA)

I INTERVALLO, $f: I \rightarrow \mathbb{R}$ CONTINUA

SE $\exists x_0, x_1 \in I$ T.C.

$$f(x_0) \cdot f(x_1) < 0$$

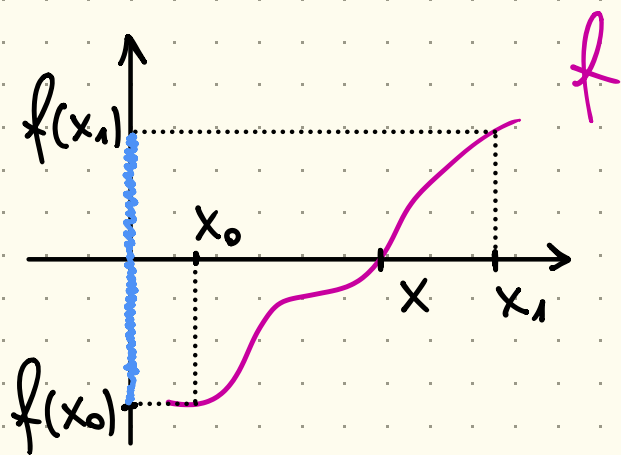
$$\begin{array}{cc} f(x_0) < 0 & f(x_0) > 0 \end{array}$$

$$\begin{array}{cc} f(x_1) > 0 & f(x_1) < 0 \end{array}$$

ALLORA $\exists x \in (x_0, x_1)$ T.C.



$$f(x) = 0$$



• METODO DI BISEZIONE

• $x^2 = 2 \rightarrow x = \pm \sqrt{2}$ ✓

• $x + \log x = 0$

$\downarrow$
 $f(x)$

• f CONTINUA

• $\lim_{x \rightarrow 0} f(x) = -\infty$, $\lim_{x \rightarrow +\infty} f(x) = +\infty$

• f CRESCENTE

